

Trail Priority Tool

User & Maintenance Guide — GGNRA

What This Tool Is

A website that ranks GGNRA trails by condition, funding, nearby projects, wildlife, hazard exposure, and more — so staff can compare trails and find where attention is most needed.

Trail and project data comes from PMIS/FMSS. Wildlife data comes live from iNaturalist. Hazard layers come from a GGNRA coastal vulnerability study. All of this loads live from ArcGIS Online (AGOL) — sensitive numbers (funding, condition) are not public, which is why sign-in is required.

Using the Tool

Sorting and filtering

- Check any box (replacement value, condition, hazard exposure, nearby species, etc.) to sort by it.
- Check several boxes to combine them into one score.
- The ↑/↓ button reverses the order — useful for finding trails with the least attention, not just the most.
- Filters below narrow the list by historic status, Trails Forever projects, or hazard type. There's also a search box.

Clicking a trail

Click any trail (list or map) to see its full stats, nearby projects, recent wildlife sightings, and — for San Francisco trails — nearby 311 complaints.

Hazard layers

Toggle any of five coastal hazard layers on the map: FEMA flood zones, flooding potential, shoreline/cliff retreat, sea-level rise, and tsunami hazard.

Species search and reroute sketching

Search any species to see every nearby sighting park-wide. The reroute tool lets you sketch a hypothetical new trail path and see what projects and wildlife are near it.

Access You'll Need to Maintain This Tool

Using the tool only requires an AGOL sign-in. Maintaining it — updating data or editing the code — requires four separate kinds of access:

AMRS account	To log into the FMSS/PFMD portal and download the project data export.
GitHub access — Trails Prioritization repository	To edit the tool's actual code (the webpage itself).
AGOL account	To access the map layers the tool reads from, and to manage the API credentials that let the update script write to those layers.
GitHub Codespaces	To run the update script in your browser — no separate account, just needs GitHub access to the repo.

Ask whoever currently maintains the tool to grant each of these before you're expected to update data or edit code.

Keeping the Data Current

Three steps. Do this whenever there's a new batch of project data — for example, quarterly.

- 1. Export the latest data from FMSS.
- 2. Run a script that converts it.
- 3. Run a script that pushes it to AGOL.

Once step 3 finishes, the live tool shows the new data automatically. Nothing else to deploy.

Step 1: Export from FMSS

PFMD portal → AMRS → Project Management → FMIS/PMIS Regional Project. Export as .xlsx — not the CSV button, which has repeatedly exported only the report's cover page. Confirm the file has real data before continuing.

Steps 2 & 3: Run the scripts

- Open the “trail-data-sync” repo on GitHub → Code → Codespaces → Create codespace on main.
- Drag in the .xlsx file.
- Open a terminal and run:

```
pip install requests openpyxl
python3 fmss_to_json.py "your-file-name.xlsx"
python3 push_to_agol.py
```

You should see “All done” at the end.

You should never need to type a password during this. The AGOL credential is stored once as a Codespaces secret and applied automatically. If you're ever asked to type one in, or see one on screen, stop — treat it like an exposed password and have it regenerated in AGOL.

If something goes wrong

“Very few components parsed”	You uploaded the cover-page-only export. Re-export as .xlsx and confirm it has real rows.
“Delete batch failed” / “not supported”	Editing isn't enabled on an AGOL layer. Layer item → Settings → Editing → enable Create/Update/Delete.
Token or “invalid client” error	The AGOL credential is missing or expired — needs the AGOL account holder to fix.
Tool shows “data failed to load”	A layer isn't shared with the whole organization. Layer item → Share → set to Organization.

Making Bigger Changes

Changing how the tool looks or works means editing its code — one webpage file in the Trails Prioritization GitHub repository.

This tool was originally built using Claude, an AI coding assistant, through plain-English conversation rather than hand-written code. That's a practical way to keep maintaining it: describe the change you want, have the AI make it, check the result. Have a second person review anything touching money, security, or sign-in before it goes live.